

GEOLOGICAL PATHWAY DIVERSITY MODEL

A Multi-Site Framework for Atmospheric Plasma Phenomena

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Abstract

Atmospheric plasma phenomena — persistent luminous orbs, recurrent earth lights, and earthquake lights — observed at geographically distributed sites worldwide have historically resisted unified physical explanation. The Geological Pathway Diversity Model (GPDM) proposes that these phenomena share a common mechanism family rooted in stress-activated electronic charge carrier physics, with site-specific expression determined by local lithology, fault geometry, tectonic stress regime, and atmospheric state. Five geological activation pathways are formally classified: (1) p-hole/peroxy charge carrier activation in mafic igneous rocks (Freund mechanism); (2) textural piezoelectric emission in crystallographically aligned quartz-bearing metamorphic and hydrothermal vein rocks; (3) triboelectric/microcracking broadband EMR from brittle fracture; (4) electrochemical/atmospheric plasma confinement driven by natural galvanic battery geometry; and (5) radon/alpha-ionization dusty plasma formation. A four-variable state equation governs output intensity and type across all pathways. Three primary field sites — Brown Mountain (North Carolina, USA), Hessdalen Valley (Norway), and La Peña de Juaica (Colombia) — anchor the multi-continent comparative framework. The model generates testable predictions including a tidal predictive calendar, GIS spatial methodology for new site identification, and a null-site falsification protocol. Deployment of a standardized three-sensor monitoring package (ULF magnetometer, air ion counter, IR thermal camera) at geologically qualified sites is proposed as the primary validation pathway.

1. Introduction

Seismic waves are the most studied expression of tectonic stress in the Earth's crust. They are not, however, the only one. A parallel body of observation — documented across centuries, from multiple continents, by investigators ranging from field naturalists to astrophysicists — records luminous phenomena that appear at persistent geographic locations, often in correlation with geological and atmospheric conditions rather than with discrete seismic events.

These phenomena present in two primary modes. The first is co-seismic: brief, intense flashes of light appearing at the Earth's surface immediately before, during, or after earthquake rupture — earthquake lights (EQLs) documented photographically during the Matsushiro swarm (Japan, 1965-67), the Saguenay event (Quebec, 1988), and the Kobe earthquake (Japan, 1995), among many

others. The second is sustained: long-duration luminous plasma orbs recurring at specific valley and ridge locations over decades, most systematically documented at Hessdalen, Norway, since 1984.

Until recently these two modes were treated as distinct phenomena requiring separate explanations. The discovery by Freund and colleagues (2002-2019) that essentially all igneous and high-grade metamorphic crustal rock contains dormant positive hole (p-hole) charge carriers — defect electrons in the oxygen anion sublattice, released by mechanical stress — provides a unifying physical substrate. When p-holes are activated in sufficient numbers, they propagate through unstressed rock at 200-300 m/sec via phonon-assisted electron hopping, accumulate at the Earth's surface, field-ionize atmospheric oxygen, and produce corona discharges accompanied by visible light emission, ULF electromagnetic pulses, infrared radiation, and air ionization cascades.

The Geological Pathway Diversity Model (GPDM) extends this foundation by proposing that the specific expression of atmospheric plasma phenomena at any given site is determined by a combination of geological activation pathway, local lithology, tectonic geometry, and environmental state variables. Different sites activate different dominant pathways or combinations thereof — producing the observed diversity of phenomenological expression from a shared physical substrate.

The model's core claim is that recurrent earth lights sites are not geologically arbitrary. They represent locations where geological conditions favor sustained charge carrier activation at sub-rupture stress levels, producing long-duration plasma phenomena rather than the brief co-seismic discharge associated with fault rupture. The same underlying physics operates at both ends of the expression spectrum.

2. Theoretical Foundation: The P-hole Mechanism

2.1 Peroxy Defects and Dormant Charge Carriers

Silicate minerals crystallizing from H₂O-laden magmas incorporate small amounts of water as hydroxyl pairs (OH⁻). During cooling through the 600-400°C window, hydroxyl pairs undergo a redox conversion in which protons steal electrons from adjacent oxygen anions, forming H₂ gas while the oxygens are oxidized from O²⁻ to O⁻, creating peroxy links of the form O₃Si-OO-SiO₃. These peroxy links — positive hole pairs (PHPs) — are electrically inactive in their bound state but constitute latent charge carrier capacity throughout the rock.

Freund (2010) established that peroxy defects are ubiquitous in the mineral world: every igneous and high-grade metamorphic rock in the Earth's crust carries a non-zero complement of peroxy links, as does every sedimentary rock containing detrital mineral grains from igneous and metamorphic sources. This universality is fundamental to the GPDM's multi-site applicability — the geological substrate for p-hole activation is not restricted to exotic rock types but is present wherever crystalline rock exists.

2.2 Stress Activation and Charge Carrier Propagation

When rocks are subjected to deviatoric stress, dislocations sweep through mineral grains. Each time a dislocation intersects a peroxy link, it momentarily breaks the O--O⁻ bond, releasing a mobile positive hole (h⁺) via a two-step process in which a neighboring O²⁻ donates an electron to the broken bond, becoming O⁻ and thus itself a mobile charge carrier. The stressed rock volume turns into a battery: h⁺ charge carriers flow laterally into unstressed rock while electrons flow downward toward the n-type conducting lower crust.

The boundary between stressed and unstressed rock acts as a Schottky barrier — passing h^+ carriers but blocking electrons. This rectifying behavior produces the unipolar magnetic pulse signature documented by Bleier et al. (2009, 2013) at the Alum Rock and Tacna earthquake preparation zones. P-holes propagate at 200-300 m/sec via phonon-assisted electron hopping (O_2 -donating an electron to adjacent O^- , advancing the hole by one atomic spacing of $\sim 2.8\text{\AA}$ at phonon frequencies of $\sim 10^{12}$ Hz). This propagation speed has been confirmed in laboratory impact experiments and in field measurements of the 8-second pulse arrival offset between two magnetometer stations 1 km apart at Alum Rock.

2.3 Surface Arrival and Atmospheric Coupling

When h^+ charge clouds reach the Earth's surface, they accumulate in a thin subsurface charge layer. The associated electric field, calculated by King & Freund (1984) using Schottky barrier theory, reaches 400,000 V/cm at a flat surface and significantly higher at topographic convex points — edges, ridges, and mountain peaks. This field exceeds the ionization threshold of air ($\sim 2 \times 10^6$ V/cm at corners), producing field ionization of O_2 molecules (positive airborne ion emission) followed by corona discharge onset when the field drives ionization avalanches. Each corona discharge produces a flash of visible light, a burst of RF emission in the kHz range, and a brief reversal of surface polarity as electron showers from the discharge neutralize the positive surface charge — after which h^+ influx from the bulk replenishes the charge, creating rapid-fire discharge sequences.

Two operating modes are distinguished. In battery mode — the dominant mode at non-seismic recurrent earth lights sites — sustained sub-threshold h^+ outflow produces continuous low-amplitude signals (ULF pulses, air ionization, IR emission) over timescales of hours to months. In plasma burst mode — operative at earthquake preparation zones approaching rupture — p-hole concentration in a confined asperity volume exceeds a critical threshold, forming a degenerate solid-state plasma that bursts outward explosively, producing co-seismic EQL flashes of the type documented at Saguenay (St-Laurent, Derr & Freund 2006) and Kobe (Tsukuda 1997).

3. The Five-Pathway Classification System

The GPDM classifies atmospheric plasma phenomena at recurrent sites into five geological activation pathways. Each pathway reflects a distinct primary mechanism of charge carrier generation or plasma formation, though pathways frequently co-occur at the same site with local geology and tectonic conditions determining which is dominant. The OR logic structure is critical: the four-variable state equation (Section 4) operates as a parallel system — a single pathway achieving threshold activation is sufficient to produce observable output. Sites where multiple pathways are simultaneously active produce the most complex and sustained phenomenology.

Pathway 1 — P-hole / Peroxy (Freund Mechanism)	
Host Geology	Mafic igneous rocks (gabbro, basalt, granite, anorthosite). PHP density highest in rocks crystallized from H_2O -laden magmas.
Trigger	Deviatoric stress — tectonic loading, S-wave shear. Dislocation movement intersects peroxy links, releasing mobile h^+ charge carriers.
Output	Pre-seismic ULF magnetic pulses; positive airborne ion emission; corona discharge at topographic highs; co-seismic EQL flash (plasma burst mode); satellite IR thermal

	anomaly; ionospheric TEC perturbation; groundwater H ₂ O ₂ formation.
Key Sites	Saguenay, Quebec (1988 M6.5 documented EQL); Kobe, Japan (1995 M7.2); Alum Rock, CA (2007 M5.4 multi-instrument); Lima, Peru (2007 M8.0); Matsushiro, Japan (1965-67).
Primary Citations	Freund 2003, 2007a, 2010, 2019; St-Laurent, Derr & Freund 2006; Bleier et al. 2009, 2013; Grant et al. 2011.

Pathway 2 — Textural Piezoelectric

Host Geology	Quartz-bearing rock with oriented electric axis polarity. Requires lattice preferred orientation PLUS polar axis alignment — specifically: hydrothermal vein quartz in fault zones; high-grade metamorphic quartzite with Leydolt/Dauphine twinning.
Trigger	Stress application along polar axis direction; shear stress on Dauphine-twinned quartzite. Condition: both crystallographic texture AND polarity orientation required for net macroscopic field.
Output	Net electric field emergence at rock surface; plasma initiation; EQL in quartz-rich geological settings; broadband EM emission during fracture (8-40 kHz dominant).
Key Sites	Kaikoura, NZ (greywacke, 2016 M7.8 documented EQL); Hessdalen, Norway (vein quartz component); Brown Mountain, NC (Appalachian metamorphic quartzite, predicted).
Primary Citations	Nikitin & Ivankina 1995; Bishop 1981; Whitehead & Ulusoy 2025; Nitsan 1977; Hanson & Rowell 1982.

Pathway 3 — Triboelectric / Microcracking

Host Geology	Any brittle quartz-bearing rock undergoing fracture. Enhanced in texturally aligned metamorphic quartzite and granite with high stress drop events.
Trigger	Brittle fracture events with discrete stress drops; microcracking; grain boundary friction. EM emission scales with stress drop magnitude (crack size), not stress level.
Output	Broadband EMR impulses concentrated below 40 kHz; directional radiation pattern; correlated with acoustic emission. Does not produce sustained plasma phenomena alone.
Key Sites	Obir Cave fault, Eastern Alps (documented ULF/LF EMR monitoring); Galena Mine quartzite, Idaho (laboratory documented); Edgar Mine, Colorado (seismic tomography validated).
Primary Citations	Baron et al. 2022; Hanson & Rowell 1982; Nitsan 1977; Rabinovitch et al. 1995, 2000, 2001.

Pathway 4 — Electrochemical / Atmospheric (Natural Battery)

Host Geology	Sites with electrochemically differentiated mineral deposits producing galvanic potential across fault geometry. Specifically: sulfide mineral oxidation/reduction couples separated by fault or valley geometry with groundwater ionic transport.
Trigger	Continuous electrochemical current driven by mineral oxidation/reduction differential. Circuit closure through electrolytic groundwater conductivity. Does NOT require seismic stress input — operates under sustained tectonic loading below fracture threshold.
Output	Long-duration (seconds to hours) thermally self-regulating plasma orbs; clustering effect; thermally stable across luminosity changes; IR + radar signatures; VLF Doppler-like

Key Sites	signals. Primary mechanism for sustained earth lights distinct from co-seismic EQL.
	Hessdalen, Norway (primary documented case — iron/zinc/copper mineral battery confirmed by VLF survey); Marfa, Texas (spectroscopic documentation).
Primary Citations	Teodorani 2004; Turner 2003; Monari et al. 2013; Vargemezis et al. 2024; Zou 1995.

Pathway 5 — Radon / Alpha-Ionization (Dusty Plasma)	
Host Geology	Uranium/thorium-bearing rock in fault zones with radon emanation pathways. Sulfide minerals with associated radioactive trace elements. Radon concentration in groundwater proportional to U concentration in adjacent aquifer rocks.
Trigger	Radon-222 decay (half-life 3.8 days) produces alpha particles → air/dust ionization (~10 ⁵ ion pairs/cm ³). Microcrack formation reduces effective grain size, releasing radon to groundwater. P-hole activity (Pathway 1) may chemically mobilize radon, coupling Pathways 1 and 5.
Output	Coulomb crystal cluster formations in dusty plasma; long-duration luminous plasmoids with helium spectral signature (alpha decay product); background ionization floor (30-300 ions/cm ³ /sec) that Pathway 1 corona discharges amplify to 10 ⁵ + ions/cm ³ .
Key Sites	Hessdalen, Norway (helium spectral signature detected; Paiva & Taft 2010 Coulomb crystal model); Kobe precursor zone (10x radon spike 9 days before M7.2).
Primary Citations	Paiva & Taft 2010; Igarashi et al. 1995; Tsytovich et al. 2007; Freund 2010 (p-hole/radon coupling).

Note on Pathway 1 / Pathway 4 Bifurcation: Both Pathway 1 (co-seismic EQL) and Pathway 4 (sustained plasma orbs) may share p-hole surface arrival as a common trigger. The bifurcation point is Variable 3 (atmospheric state): dry conditions with high topographic relief favor corona discharge and EQL flash; humid valley conditions with electrochemical confinement mechanism favor thermally self-regulating plasma orb formation. Brown Mountain (dry Appalachian ridges) is predicted to be a predominantly Pathway 1/2 output site; Hessdalen (humid Norwegian valley with natural battery geometry) is a predominantly Pathway 4 output site. The same fundamental charge carrier physics underlies both.

4. The Four-Variable State Equation

The GPDM's output intensity and type at any given site and moment is governed by four primary variables. The equation is not a simple linear function — it operates with OR logic across pathways and has a branching structure at Variable 3:

$$\Phi(\text{output}) = f(V1, V2, V3, V4)$$

Variable 1 — Stress Rate (dσ/dt)

Controls the rate of dislocation generation and therefore the rate of PHP dissociation and h• production. Stress input sources are multiple: tectonic loading (primary for co-seismic EQL), S-wave shear passage (documented trigger, Freund 2019), tidal loading (Baron et al. 2022 confirm tidal

ground surface displacement of up to 10cm at monitoring sites), and microseismic background from distant ocean wave interactions (Bromirski 2001). Critical finding from Freund (2010): p-hole carriers persist for days to months under sustained constant stress, with three distinct carrier lifetime populations. This means tidal cycles modulate the output intensity of an already-active system rather than switching it on and off discretely. Variable 1 can be near-zero at Hessdalen (seismicity below M2.5 since 2000) while Pathway 4 continues operating — confirming that Variable 1 is not required for all pathways.

Variable 2 — Carrier Density / Lithology (ρ_{PHP})

The PHP concentration in the geological substrate, which is a function of rock type and crystallization history. Highest in mafic igneous rocks (gabbro, basalt) crystallized from H₂O-laden magmas at depth. Also present in all igneous and metamorphic crustal rock per Freund's universality statement — the question is magnitude rather than presence. For Pathway 2, Variable 2 encompasses both quartz content and the presence of oriented electric axis polarity (the Nikitin & Ivankina (1995) condition): lattice preferred orientation alone is insufficient — polarity orientation is required for net macroscopic piezoelectric field emergence.

Variable 3 — Atmospheric / Environmental State (Ψ_{atm})

The bifurcation variable. Controls the output type given that charge carriers have reached the surface. High humidity (80-90%, documented at Hessdalen during events by Teodorani 2004) enables electrochemical confinement via Turner's thermochemical refrigeration mechanism, producing sustained plasma orbs (Pathway 4 output). Low humidity or high topographic relief with discrete convex geometry favors corona discharge and EQL flash (Pathway 1/2 output). Also governs the efficiency of Pathway 5 (radon/dusty plasma): water vapor density affects Coulomb crystal formation in alpha-ionized dusty plasma. The winter dominance and humidity correlation of Hessdalen events (Strand 1984, Teodorani 2004) is a direct expression of Variable 3.

Variable 4 — Geometric Confinement and Circuit Closure Pathway

The structural geometry of the charge carrier activation volume and its surface emergence pathway. Includes: dyke and fault orientation relative to stress field (governing $h \bullet$ flow direction); topographic geometry at emergence point (governing electric field concentration); and circuit closure mechanism — three natural pathways identified by Freund (2010): (i) downward electron flow to the n-type conducting lower crust at ~500°C; (ii) electrolytic circuit closure through saline groundwater; (iii) atmospheric circuit closure via massive air ionization. The Hessdalen natural battery geometry (Monari et al. 2013, Vargomezis et al. 2024) is a specific expression of Variable 4 in which the electrochemical potential difference between spatially separated mineral deposits drives a circuit with groundwater as the electrolyte.

5. Primary Field Sites

Three primary field sites anchor the GPDM's comparative multi-site framework. Each site has independent documentation of recurrent atmospheric plasma phenomena, geological characterization that assigns it to one or more GPDM pathways, and an active research relationship supporting future monitoring deployment.

Site	Geology	Phenomena	Active Pathways	Status
Brown Mountain, NC, USA	Appalachian fold-thrust belt; metamorphic quartzite with deformation texture (Leydolt/Dauphine twinning); hydrothermal vein quartz in fault zones; mixed PHP-bearing metamorphic rock.	Recurrent luminous orbs documented since 19th century. Events concentrated at ridges and topographic highs — consistent with p-hole corona discharge geometry (Freund 2003).	1 (primary), 2 (active — textured quartzite), 3 (microcracking)	First formal GPDM field site writeup planned. Outreach to Appalachian State University researcher sent.
Hessdalen Valley, Norway	Gabbro and amphibolite formations (PHP-bearing, Freund mechanism substrate); quartz-rich mica schists; sulfide mineral deposits (iron/zinc west, copper east); VLF-confirmed conductive zones tracing 6x12 km ellipse of gabbro intrusion.	Most instrumented earth lights site on Earth. 40 years of documented data. Key anomalies: flat-topped steep-sided luminosity distribution; thermally self-regulated plasma orbs; clustering effect; VLF Doppler signals; radar tracks.	4 (primary — natural battery confirmed by VLF survey); 1 (substrate present, seismicity insufficient for activation); 2 (vein quartz component); 5 (radon/dusty plasma)	Active collaboration pathway. GPDM v2 sent to Fred Pallesen (CEO, Project Hessdalen). Virtual status meeting attended.
La Pena de Juaica, Colombia	Andean setting; geological characterization in development.	Recurrent atmospheric lights documented. Southern hemisphere anchor site for multi-continent comparison.	To be determined pending geological literature	Framework anchor site. Geological literature search in progress.

Note on Hessdalen Pathway Assignment: Vargemezis et al. (2024) confirmed that Hessdalen valley seismicity has not exceeded M2.5 since 2000, explicitly ruling out Freund's seismic stress trigger (Pathway 1) as the primary active mechanism. The gabbro substrate is present (PHP-bearing), but the stress trigger is insufficient. Pathway 4 (natural battery, confirmed by VLF survey and EFM measurements) is the primary active mechanism. This finding strengthens the GPDM by demonstrating that Variable 1 (stress rate) can be near-zero while the system remains active through Pathway 4's electrochemical circuit — confirming the OR logic structure of the state equation.

6. The Tidal Predictive Calendar

The GPDM's tidal predictive calendar is grounded in the confirmed relationship between tidal loading and rock strain at monitoring sites. Baron et al. (2022) documented tidal ground surface displacement of up to 10cm vertical at the Obir Cave fault monitoring site using the Goddard GOT00.2 ocean tide model, establishing that tidal loading cycles produce measurable strain changes at geologically active sites.

The critical revision from Freund (2010) is that the tidal calendar does not predict discrete activation events — it predicts output intensity modulation of an already-active system. P-hole carriers persist for days to months under sustained tectonic stress (three distinct lifetime populations documented in gabbro at 48 MPa). Tidal loading cycles therefore modulate the surface-arriving h• flux rather than switching carrier generation on and off. The calendar predicts: enhanced corona discharge probability during tidal stress maxima; suppressed output during tidal unloading phases; and compounded enhancement during coincident tidal stress maxima and elevated regional seismicity.

For Brown Mountain (Appalachian location), tidal prediction must account for both solid earth tides and the microseismic background from Atlantic Ocean wave activity (Bromirski 2001 established transcontinental propagation of microseismic energy from coastal wave-wave interactions). The dominant microseismic source areas — near southern Massachusetts coast and south of Cape Hatteras — are within Rayleigh wave propagation range of the Appalachians. This provides a continuous low-amplitude stress input to Brown Mountain's geological system that is modulated by ocean storm state independent of tidal cycles.

Specific tidal predictions for Brown Mountain and Hessdalen require site-specific tidal loading models. This is identified as a primary next development step for GPDM v3.

7. GIS Spatial Prediction Methodology

The GPDM's GIS spatial prediction methodology maps the geological conditions required for each pathway to identify high-probability new sites where atmospheric plasma phenomena should occur but have not yet been systematically documented.

The primary mapping layers are: (1) mafic igneous and high-grade metamorphic rock distribution (Pathway 1 substrate); (2) fault zone vein quartz and metamorphic quartzite with deformation texture (Pathway 2 substrate); (3) electrochemically differentiated mineral deposit pairs separated by fault geometry with groundwater connectivity (Pathway 4 substrate); (4) uranium/thorium-bearing rock in fault zones with documented radon emanation (Pathway 5 substrate).

The Vargemezis et al. (2024) conclusion explicitly invites this methodology: 'The particular geological structure detected in Hessdalen valley may encourage similar campaigns in other areas where similar phenomena are observed.' The GPDM formalizes the geological selection criteria that would guide such campaigns.

Two forward site predictions have been made under the GPDM framework based on this methodology. These predictions will be formally documented in a companion technical note and submitted for independent geological assessment prior to field verification.

8. Null-Site Falsification Protocol

A scientific framework that cannot be falsified is not science. The GPDM's null-site falsification protocol defines the conditions under which the framework's predictions are demonstrably wrong.

The protocol identifies geologically equivalent sites — matching the lithology, fault geometry, tectonic stress regime, and atmospheric conditions of confirmed GPDM sites — where no atmospheric plasma phenomena have been documented. If the GPDM's pathway conditions are both necessary and sufficient predictors of phenomena, null sites should either: (a) lack one or more required pathway variables; or (b) represent sites where phenomena occur but have not been reported due to low population density or observer access.

The protocol design: Select five candidate null sites matched to Brown Mountain's Appalachian quartzite geology. Assess each against the four-variable state equation. Predict which should show phenomena (positive prediction) and which should not (null prediction). Deploy the standard three-sensor monitoring package at all five for a minimum 12-month period. If positive-predicted sites show no signal and null-predicted sites show signal, the framework is falsified. If results match predictions, the framework is strengthened.

The null-site test is explicitly designed to be run in parallel with the monitoring network deployment proposed in Section 9. It converts the deployment from a descriptive monitoring exercise into a controlled scientific test.

9. Proposed Monitoring Network

The validation pathway for the GPDM is a standardized multi-instrument monitoring network deployed at geologically qualified sites. The instrument package is derived from the QuakeFinder/Freund methodology validated at the Alum Rock M5.4 earthquake (Bleier et al. 2009) and the Bass Lake granite boulder field experiment (Bleier et al. 2013):

Three-Sensor Standard Package: (1) ULF induction magnetometer (0.01-12 Hz, sensitivity ~0.1 pT/√Hz at 1 Hz) for unipolar pulse detection; (2) Air ion counter / conductivity sensor for positive and negative airborne ion monitoring; (3) IR thermal camera (8-12 μm) for non-thermal IR emission and night-time cooling slope analysis.

Optional Enhancement: (4) Radon monitor in groundwater or soil gas; (5) Geophone for acoustic emission correlation; (6) GPS-synchronized timing for multi-site pulse propagation analysis.

The network design prioritizes three anchor deployments as Phase 1: Brown Mountain, NC (Western hemisphere anchor); Hessdalen Valley, Norway (expansion of existing AMS station, international precedent); and one Canadian Shield site within the Western Quebec Seismic Zone (new deployment, dual-use UAP/seismic precursor mandate).

Phase 2 expands to additional NATO member-state sites with confirmed geological qualification: Finland (Paasselkä), Italy (Monti Sibillini), Greece (Eastern Mediterranean seismic zone), and Turkey (North Anatolian Fault zone). Standardized instrument package and data protocol enables the first interoperable multi-continent atmospheric plasma phenomena dataset.

The Canadian and NATO deployment framing leverages the network's dual scientific mandate: UAP characterization (standardized scientific monitoring of anomalous aerial phenomena) and seismic precursor research (pre-signal monitoring using the validated Freund/Bleier methodology). One network, one dataset, two government mandates served.

10. Conclusions and Forward Predictions

The Geological Pathway Diversity Model establishes that recurrent atmospheric plasma phenomena at persistent global sites are not geographically arbitrary and not mechanistically diverse. They represent the surface expression of a family of geological charge carrier activation processes sharing a common physical substrate — peroxy defect chemistry in crustal silicate minerals — with site-specific expression determined by local lithology, fault geometry, tectonic stress regime, and atmospheric state.

Five activation pathways are formally classified with peer-reviewed citation support across all primary claims. The framework generates testable predictions at three levels: site-specific (tidal calendar predictions for Brown Mountain and Hessdalen); inter-site (geological matching predictions for new candidate sites); and null-site (falsification predictions for geologically non-qualifying sites).

The framework is novel in two respects. First, it explicitly connects the phenomenon at non-seismic recurrent earth lights sites (Hessdalen, Brown Mountain) to the pre-earthquake signal literature (Freund, Bleier, QuakeFinder) through the shared p-hole charge carrier mechanism — a connection that has been implicit in the literature but not formally articulated as a classification framework. Second, it proposes a standardized monitoring methodology that would generate the first interoperable multi-continent dataset for these phenomena, enabling genuine scientific hypothesis testing rather than continued case-by-case observation.

The GPDM's forward predictions: (1) Tidal stress maxima will correlate with increased event frequency at Brown Mountain and Hessdalen at statistically significant levels over a 24-month monitoring period. (2) Two forward site predictions made under the GIS spatial prediction methodology will show anomalous atmospheric phenomena upon standardized monitoring deployment. (3) The null-site falsification protocol will correctly classify at least 4 of 5 candidate sites.

These predictions are falsifiable, instrumentally testable, and grounded in a peer-reviewed scientific foundation. The monitoring network deployment is the next step.

11. Citation Status

The following citations have been verified as part of the GPDM evidence base. Full text has been obtained and analyzed for all citations marked as verified.

Citation	Title	Journal / Source	Status
Freund, F.T. (2003)	Rocks That Crackle and Sparkle and Glow	J. Scientific Exploration 17(1): 37-71	Full text verified
Freund, F.T. (2007a)	Pre-earthquake signals Part I	NHESS 7: 535-541	Full text verified
Freund, F.T. (2007b)	Pre-earthquake signals Part II	NHESS 7: 543-548	Full text verified
Freund, F.T. (2010)	Toward a Unified Solid State Theory	Acta Geophysica 58: 719-766	Full text verified
Freund, F.T. (2019)	Co-seismic Earthquake Lights	Pure & Applied	Full text verified

		Geophysics 176: 3439-3450	
Freund, Takeuchi & Lau (2006)	Electric currents from stressed igneous rocks	Phys. Chem. Earth 31: 389-396	Confirmed real
St-Laurent, Derr & Freund (2006)	Earthquake Lights and P-hole Carriers	Conference proceedings	Full text verified
Bleier et al. (2009)	ULF, air conductivity, IR — Alum Rock M5.4	NHESS 9: 585-603	Full text verified
Bleier et al. (2013)	Ground-based and space-based EM monitoring	TERRAPUB chapter	Full text verified
Grant et al. (2011)	Ground Water Chemistry & Animal Behavior	IJERPH 8: 1936-1956	Full text verified
Balk et al. (2009)	Oxidation of water to H2O2 at rock-water interface	EPSL 283: 87-92	Confirmed real
Baron et al. (2022)	EMR at seismogenic fault — Obir Cave	Engineering Geology 311: 106912	Full text verified
Teodorani, M. (2004)	Long-Term Scientific Survey of Hessdalen	J. Scientific Exploration 18(2): 217-251	Full text verified
Teodorani, M. (2024)	Investigating UAP Using Astronomical Techniques	Limina 1(1): 40-54	Full text verified
Vargemezis et al. (2024)	VLF survey — Hessdalen lights	J. Applied Geophysics 226: 105398	Full text verified
Whitehead & Ulusoy (2025)	EQL: Rayleigh Scattering, Kaikoura	Atmosphere 16(3): 277	Full text verified — open access
Igarashi et al. (1995)	Ground Water Radon — Kobe earthquake	Science 269: 60-64	Full text verified
Nikitin & Ivankina (1995)	Mechanisms of Piezoelectric Active Rocks	Textures & Microstructures 25: 33-43	Full text verified
Bishop, J.R. (1981)	Piezoelectric effects in quartz-rich rocks	Tectonophysics 77: 297-321	Confirmed real
Nitsan, U. (1977)	EM emission from quartz-bearing rock fracture	Geophys. Res. Lett. 4: 333-336	Confirmed real
Hanson & Rowell (1982)	EM Radiation from Rock Failure	USBM RI 8594	Full text verified
Paiva & Taft (2010)	Dusty plasma mechanism — Hessdalen lights	JASTP 72: 1200-1203	Confirmed real
Tsyтовich et al. (2007)	Plasma crystals to inorganic living matter	New J. Physics 9: 263	Confirmed real
Turner, D.J. (2003)	The missing science of ball lightning	J. Scientific Exploration 17(3): 435-496	Confirmed real
Monari et al. (2013)	Hessdalen — A Perfect Natural Battery	ICPH Articles	Confirmed real
Strand, E.P. (1984)	Project Hessdalen Final Technical Report	Project Hessdalen	Full text verified
Hauge, B.G. (2010)	Transient luminous phenomena	Acta Astronautica	Confirmed real

	— Hessdalen		
Derr, J.S. (1973)	Earthquake lights: review of observations	BSSA 63: 2177-2187	Confirmed real
Derr, J.S. (1986)	Luminous phenomena and rock fracture	Nature 321: 470-471	Confirmed real
St-Laurent, F. (2000)	Saguenay earthquake lights 1988-1989	Seismolog. Res. Lett. 71: 160-174	Confirmed real

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